

Compliance for Free: Learning Identifiable Impedance via Bilateral Teleoperation

Anonymous Authors

Abstract—Vision Language Models often generate positions for task completion. However, contact-rich manipulation requires impedance. In this paper, we extract compliance without any dedicated force-torque sensor, but using only the torque sensing already present on the manipulator. Recent attempts to close this gap treat force as an additional observation fed into the policy; the handful that produce force outputs must either assume a known task structure or infer compliance targets from simulation with privileged contact state, because pose-only teleoperation (VR controllers, SpaceMouse, UMI grippers) does not measure what impedance the human was regulating. Four-channel bilateral control does. We show that the leader-arm trajectory is an independent measurement of the operator’s intended equilibrium pose, which uniquely resolves the equilibrium/stiffness identifiability ambiguity that has forced prior work into simulation. This yields per-timestep, anisotropic, real-world compliance labels at zero additional annotation cost. We use these labels to supervise a compliance head on a pretrained VLA.

Index Terms—component, formatting, style, styling, insert.

I. INTRODUCTION

Vision-Language-Action (VLA) models have revolutionized robot manipulation [1]–[5], yet they consistently fail at the last millimetre. Contact-rich manipulation tasks such as wiping, tight tolerance insertion, and assembly require a robot to modulate how hard a robot pushes rather than just where it goes. In these scenarios, compliance is a first class action, every bit as critical as the target pose itself.

The robotics community remains trapped in a data bottleneck regarding this physical interaction. Recent models have attempted to close the gap by injecting contact force as an additional input observation via a force-aware mixture of experts [6] or hybrid control frameworks [7]. However, the few architectures that actively output force or compliance targets still lean heavily on structured constraints. The few architectures that actively output force or compliance targets still lean heavily on structured constraints. They require hand-specified interaction frames [8], infer stiffness matrices from privileged simulation contact data [9], or distill policies from dedicated inference-time tactile hardware [10].

This barrier is not a failure of model architecture, but a fundamental mathematical flaw in the demonstration interfaces. The standard setups driving modern visuomotor policy learning such as low-cost bimanual rigs [11], universal manipulation interfaces [12], or VR controllers [13] record only the robot’s realized pose. Because pose and force alone leave the operator’s intended equilibrium and stiffness confounded,

these interfaces suffer from a strict identifiability problem. They have never measured the impedance the human intended, and mathematically, they never will.

To resolve this ambiguity, we turn to four-channel bilateral control [14]. Unlike passive recording interfaces, a bilateral setup actively couples a leader and follower arm. We demonstrate that the leader-arm trajectory serves as an independent, physical measurement of the operator’s intended equilibrium pose. By explicitly measuring what the human operator commands rather than just where the robot ends up, we collapse the equilibrium and stiffness ambiguity that has forced prior work into simulation. Critically, this resolution allows us to extract compliance without relying on dedicated force-torque or tactile sensors. Utilizing only the joint-torque sensing already present on the manipulator, we introduce a masked extraction procedure that yields per-timestep, anisotropic, real-world compliance labels. We subsequently use these labels to supervise a compliance head on a pretrained VLA backbone.

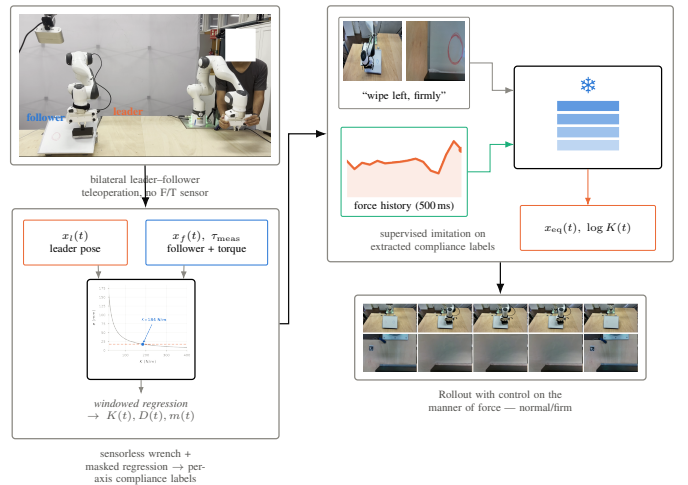


Fig. 1: From bilateral demonstrations to instruction-conditioned compliance.

In summary, this paper makes the following contributions:

- A sensorless extraction procedure that resolves the pose-only identifiability ambiguity via bilateral teleoperation, yielding per-timestep compliance labels using only native joint-torque sensors.
- An offline benchmark showing that the labels carry learnable signal: a trained predictor beats a per-axis constant predictor on every translational axis, for a $1.25\times$ aggregate improvement.

- A compliance-output VLA policy trained on these extracted labels that safely tracks chunked impedance targets and is the only policy of five whose realized contact force responds to the manner instruction.

The rest of the paper is organized as follows. Section II...Section III formalizes the identifiability argument, the extraction procedure, the sensorless wrench estimator that makes it possible without dedicated force hardware, the compliance-output policy, and the low-level controller that tracks its output safely. Section IV reports what this produces on real hardware: dataset statistics, the offline benchmark, and the real-robot policy comparison.

II. RELATED WORK

A. VLA with force and similar

A large-scale of vision-language-action (VLA) policies such as RT-1 [1], RT-2 [2], Octo [3], π_0 [4], and SmolVLA [5] learn a generalist policy mapping from vision and language to robot actions at scale. However, each of them only outputs pose, joint, or gripper targets, the stiffness for the low-level controller is fixed, not usually predicted by the policy.

Recent attempts to close this gap have moved beyond fixed stiffness. Models like ForceVLA [6] and FD-VLA [15] incorporate contact force as an input feature, either via mixture-of-experts or cross-modal distillation, to make position outputs ‘force-aware’. Meanwhile, hybrid frameworks like ForceVLA2 [7] predict target forces alongside position. However, these methods either treat force as a rigid target rather than variable compliance, or require dedicated F/T sensors during training and inference.

Among policies that produce compliance without F/T hardware, each leans on structure this paper’s setting lacks. Force Policy [8] relies on a hand-specified interaction frame. Stiffness Copilot [9] infers direction-dependent stiffness matrices in simulation using privileged contact information. And recent approaches like PaCo-VLA [16] structure compliance via hand-designed task stages and semantic bindings. Even though, HapticVLA [10] removes tactile sensing at inference by distilling from real tactile supervision collected at training time, it needs a tactile sensor for training. Our method bypasses these constraints by extracting pure, unrestricted impedance targets directly from human demonstrations.

Bi-LAT [17] is closest in spirit: it also learns from bilateral-control demonstrations and modulates applied force by language (“softly grasp” vs. “strongly twist”), but it regresses joint torque directly rather than extracting an explicit, physically interpretable stiffness label, so its notion of force control does not rest on an identifiability argument and cannot be reused to supervise a different backbone or task. None of these five requires zero dedicated force/tactile hardware at both training and inference time, and none derives compliance supervision from an identifiability argument applicable to any existing bilateral-teleoperation dataset, the gap this paper’s extraction procedure (Section III-B) and policy (Section III-F) fill. Closest to our setting, Xu et al. [18] also read impedance

out of a bilateral leader–follower rig without force sensing, treating the mismatch between leader and follower response as an implicit encoding of interaction force, and cloning operator *intent* rather than realized motion. Their impedance remains implicit, no stiffness label is ever produced.

B. Demonstration interfaces for policy learning

The pose-only interfaces constrained by our identifiability argument are precisely those generating the vast majority of VLA training data. The Universal Manipulation Interface [12] collects demonstrations with a handheld gripper that records only its own trajectory. ALOHA’s low-cost leader-follower hardware [11] rely entirely on rigid pose tracking. Whether using UMI, ALOHA, VR controllers, or SpaceMouse setups, these systems only record where the demonstrator moved the robot, none records what stiffness the demonstrator intended, effectively bottlenecking the extraction of compliance labels. Our method bridges this gap by replacing passive pose-only recording with active bilateral coupling, supplying the independent measurement these interfaces lack.

C. Bilateral Teleoperation and Control

Four-channel bilateral teleoperation is a mature control technique, with stability and transparency thoroughly analyzed since Lawrence [14]. We leverage this theoretical maturity rather than developing bilateral control itself as a core contribution.

Recent work has begun exploring bilateral setups for imitation learning data collection. Yamane et al. [19] build a sensorless 4-channel framework using disturbance observers to estimate force for manipulators lacking joint-torque sensors, while Inami et al. [20] exploit position/force duality to retroactively edit recorded motions. Prometheus [21] introduces a mocap-based teleoperation system, but requires adding a dedicated compression force sensor at the gripper for operator feedback.

Our approach diverges from this recent literature in two fundamental ways:

- Unlike Prometheus [21], our wrench estimation and compliance labels utilize only the manipulator’s native joint-torque sensors, requiring zero additional force/torque hardware.
- Unlike Yamane et al. [19], we do not merely estimate a scalar external force. We extract an explicit, interpretable per-axis stiffness matrix to supervise a compliance-output VLA policy.

III. METHODOLOGY

A. The Identifiability Problem

We model contact interaction at the end-effector with a diagonal, time-varying Cartesian impedance law:

$$\mathbf{f}(t) = \mathbf{K}(t) \odot \mathbf{e}(t) + \mathbf{D}(t) \odot \dot{\mathbf{e}}(t), \quad \mathbf{e}(t) = \mathbf{x}_{\text{eq}}(t) \ominus \mathbf{x}_f(t), \quad (1)$$

where $\mathbf{x}_f(t)$ is the measured follower pose, $\mathbf{x}_{\text{eq}}(t)$ is the operator’s (unobserved) commanded equilibrium pose, $\mathbf{f}(t)$ is the measured contact wrench, and $\mathbf{K}(t), \mathbf{D}(t) \in \mathbb{R}^6$ are diagonal stiffness and damping vectors expressed in the contact

frame. The difference operator $\ominus$ is Euclidean subtraction on the translational channels and the log map on $SO(3)$ for the rotation. $\odot$ denotes element-wise product.

Given only the pair $(\mathbf{x}_f, \mathbf{f})$ in dataset, Eq. (1) is *unidentifiable*: the equation $f_i = K_i(x_{eq}^i - x_f^i) + d_i(\dot{x}_{eq}^i - \dot{x}_f^i)$ for any axis i and any $k_i(t) > 0$, there exists a compatible $x_{eq}^i(t)$ (and a compatible $d_i(t)$) that reproduces the observed $f_i(t)$ exactly. Pose-only teleoperation interfaces mathematically cannot resolve this ambiguity because they fail to record an independent measurement of the intended equilibrium $(x_{eq}^i(t))$.

B. Resolving Identifiability via Bilateral Teleoperation

Four-channel bilateral teleoperation fundamentally alters this paradigm. By actively coupling a leader and follower arm, the system provides a direct measurement of the operator's intent.

$$x_{eq}(t) := x_l(t) \text{ (leader pose, independently measured)} \quad (2)$$

Substituting the leader pose collapses the ambiguity: $e(t) = x_l(t) \ominus x_f(t)$ is now directly observed, and $(K(t), D(t))$ become identifiable by regression on $(e, \dot{e}, f)$, subject to sufficient excitation in the data.

C. Compliance label extraction

Contact-frame definition: We fit a contact frame for each demonstration, or pooled across a session when the physical contact surface is fixed throughout. Given a set of in-contact follower positions $\mathbf{p}_j \in \mathbb{R}^3$, the centered position covariance and its singular value decomposition is calculated as follows,

$$[U, \Sigma, V^\top] = \text{SVD}(\{p_j - \bar{P}\}), \quad (3)$$

$$\hat{\mathbf{n}} = \mathbf{v}_3, \quad \text{planarity_ratio} = \sigma_3 / \sigma_1, \quad (4)$$

where, v_3 is the right singular vector associated with the smallest singular value σ_3 . A fit is reject when planarity_ratio > 0.15 .

Windowed regression for (k, d) : For each axis i , we estimate (k_i, d_i) over a sliding 300ms window directly at the 1kHz control rate by solving a regularized, box-constrained nonlinear least-squares problem:

$$\min_{k_i, d_i} \sum_{t \in W} (f_i(t) - k_i e_i(t) - d_i \dot{e}_i(t))^2 + \lambda \|\log k_i - \log k_i^{\text{prior}}\|^2 \quad (5)$$

subject to $k_i \in [k_{\min}, k_{\max}]$ and $d_i \geq 0$. Because the regularizer is nonlinear in k_i , this does not admit a closed-form ridge solution and is solved in log-space. The bounds $[k_{\min}, k_{\max}]$ are strictly defined by the realistic safety limits of the physical Cartesian impedance controller. We use critically damped $\mathbf{D}$ values derived from $\mathbf{K}$ ($D_i = 2\sqrt{K_i}$).

Identifiability mask: For each axis i and timestep t , a binary validity mask is defined as follows,

$$m_i(t) = \mathbb{I}[\text{cond}(J_W) < \kappa_{\max}] \cdot \mathbb{I}[|e_i(t)| > \sigma_e] \cdot \mathbb{I}[|f_i(t)| > \sigma_{f,i}] \cdot \mathbb{I}[\text{contact}(t)], \quad (6)$$

combining a regression-conditioning term, a minimum-displacement excitation term, a per-axis force-above-noise-floor term (from $\sigma_{f,i}$), and a contact indicator. Figure 3b shows per-axis mask coverage, computed within in-contact timesteps excluding free-space and reset time motion.

D. Wrench Estimation

At static poses, ($\dot{q} = \ddot{q} = 0$), the residual between measured joint torque and the arm's zero-payload gravity model is linear in the unknown payload's mass and mass-weighted center of mass:

$$\Delta\tau(q) = \tau_{\text{meas}}(q) - \tau_{\text{grav},0}(q) = J_p(q)^\top \begin{bmatrix} m \\ mc_x \\ mc_y \\ mc_z \end{bmatrix}, \quad (7)$$

which we solve by least squares over ≥ 8 static poses. This is the same identifiability logic as Sections III-A, III-B.

Immediately before every session we sample the wrench bias at a fixed free-space pose and subtract it as a per-axis constant offset for the remainder of the session. We fit a per-axis residual bias model over free-space sweep data with a random-Fourier feature ridge regression, a finite-dimensional closed-form approximation to Gaussian-process kernel-regression,

$$\varphi(x) = \sqrt{\frac{2}{D_f}} \left[\cos(\omega_1^\top x + b_1), \dots, \cos(\omega_{D_f}^\top x + b_{D_f}) \right]^\top, \quad (8)$$

$$\omega_k \sim \mathcal{N}(0, \ell^{-2}I), \quad b_k \sim \mathcal{U}(0, 2\pi), \quad (9)$$

$$f_{\text{bias}}(x) \approx W\varphi(x), \quad W = \underset{W}{\text{argmin}} \|F - W\Phi\|^2 + \lambda \|W\|^2, \quad (10)$$

with the closed-form ridge solution for W , fit independently per axis on $x = (q, \dot{q}) \in \mathbb{R}^{14}$. Hyperparameters for the residual bias model are reported in the Table II.

Because estimation error is strongly configuration-dependent, we pin a fixed nullspace posture and constrain the working volume during both calibration and demonstration collection.

We define the per-axis noise floor, evaluated on held-out free-space sweep sessions, as

$$\sigma_{f,i} = \min(\text{RMS}_i^{\text{raw}}, \text{RMS}_i^{\text{bias-corrected}}), \quad (11)$$

i.e. the bias-corrected estimate is used only on axes where it measurably improves over the raw estimate. $\sigma_{f,i}$ is consumed directly by mask condition (iii) in Eq. (6).

E. Label-quality validation: Offline compliance prediction benchmark

This benchmark tests whether the extracted labels carry any meaningful signal.

Task. Predict $\log K_i(t)$ only at masked, identifiable timesteps, from a feature vector that deliberately excludes force:

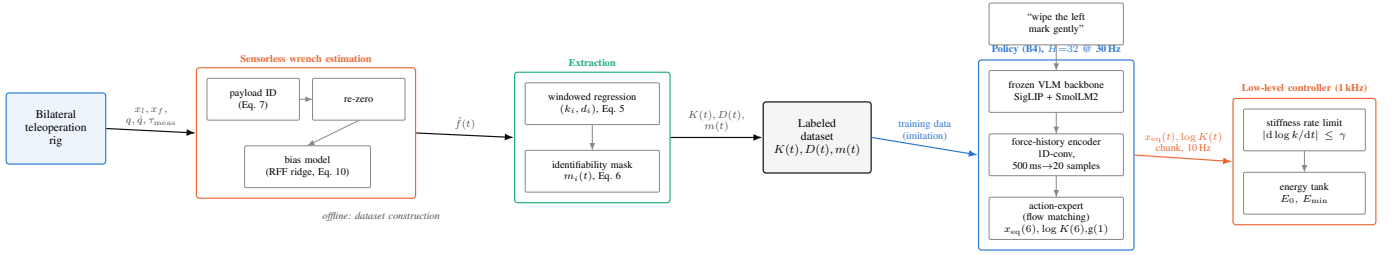


Fig. 2: System overview. Offline (left): bilateral rig $\rightarrow$ sensorless wrench estimation (payload ID $\rightarrow$ re-zero $\rightarrow$ bias model) $\rightarrow$ extraction (windowed regression + identifiability mask) $\rightarrow$ labeled dataset. Online (right): the labeled dataset trains the compliance-VLA policy (frozen VLM backbone, force-history encoder, flow-matching action expert), whose chunked $x_{eq}(t), \log K(t)$ output is tracked by a 1 kHz Cartesian impedance controller with stiffness rate-limiting and an energy tank; the dashed return arrow shows the controller driving the same follower arm used for data collection. [TODO: photo of the rig]

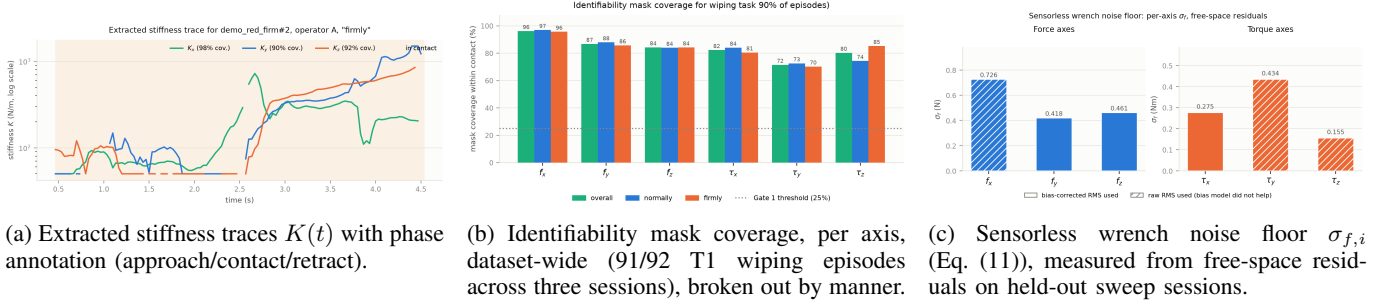


Fig. 3: Dataset validity, dataset-wide. Together these establish that the extracted compliance labels are identifiable with real margin (a), retained at high coverage after masking (b, Table I), and measured against a real, held-out noise floor well below the labels' working range rather than assumed or hand-tuned (c). Panel (c): hatched bars mark axes where the bias-corrected estimate did not improve over the raw RMS (f_x, τ_y, τ_z), for which $\sigma_{f,i}$ is the raw value; this is a free-space noise floor.

TABLE I: Per-axis mask coverage (in-contact timesteps only).

Axis	f_x	f_y	f_z	τ_x	τ_y	τ_z
Coverage (%)	96.2	86.8	84.3	82.3	71.5	80.3

TABLE II: Residual bias model hyperparameters (Eq. (8)).

Parameter	Value
Random features D_f	512
Length scale ℓ	30
Ridge penalty λ	0.1

$$x_t = [q_t \in \mathbb{R}^7, \dot{q}_t \in \mathbb{R}^7, \text{manner} \in \{0, 1\}^2, \text{referent} \in \{0, 1\}^2] \in \mathbb{R}^{20}. \quad (12)$$

Baselines.

- *Constant*: per-axis training-split mean of $\log k_i$.
- *Learned*: the same random-Fourier-feature ridge model of Eqs. (8)–(10), reused here for a second, distinct regression target.

Metric. Per axis and aggregated:

$$\text{RMSE}_i = \sqrt{\text{mean}_{t \in \text{masked test set}} (\log \hat{k}_i(t) - \log k_i(t))^2},$$

$$\text{aggregate} = \text{mean}_i \text{RMSE}_i. \quad (13)$$

F. Compliance-output Policy

Backbone. SmolVLA [5] ($\sim 450\text{M}$ parameters from LeRobot), with the pretrained vision-language model frozen and the action expert finetuned.

Inputs. We utilize two $224 \times 224 \times 3$ RGB streams: a scene camera capturing the global workspace and a rigidly mounted wrist camera. A language instruction of “wipe the {referent} mark {manner}”, where referent $\in \{\text{left}, \text{right}\}$ and manner $\in \{\text{normally}, \text{firmly}\}$. Here, the referent is spatial deixis rather than mark color, this approach is adopted as the red and blue color marks collected during the demonstrations occupied low pixel space and the model could not distinguish between them. A force-history token, a trailing 500ms of 6-DoF wrench downsampled to 20 samples (encoded by a 1D-convolutional network) is injected post-VLM (following the ForceVLA [6]).

Output. An action chunk of $H = 32$ of $[x_{eq}(6), \log K(6)]$. $x_{eq}(6)$ is absolute target pose in the same base frame convention as x_f . Absolute target pose is chosen as the identifiability argument of this paper depends on x_{eq} being directly comparable to an absolute pose.

Loss. Flow-matching (L1) on x_{eq} and masked Huber loss

on $\log K$, at a weight tuned on the validation split.

Augmentation. Force dropout ($p = 0.15$) and random wrench-bias injection (± 1 N) during training, to guard against modal masking, a failure mode in which the high dimensional visual stream dominates optimization and low dimensional force is silently ignored despite being available during training.

G. Low-level controller

Franka Research 3 manipulator is driven by a 1kHz Cartesian impedance controller,

$$\tau = J^\top [K_t(x_{\text{eq}} - x) + D_t(\dot{x}_{\text{eq}} - \dot{x})] + C(q, \dot{q})\dot{q} + g(q) + \tau_{\text{null}}, \quad (14)$$

The controller tracks the policy’s chunked $(x_{\text{eq}}(t), K(t))$ output at 1 kHz while the policy replans at ~ 10 Hz (taking 3 actions per chunk to execute).

Temporal ensembling. A chunk inferred by one forward pass and the next do not need to have a smooth transition. So executing each newly queried chunk by discarding the previous one creates a discontinuity that a rigid, non-compliant policy has nothing to physically absorb and stalls the trajectory execution. We combine every buffered chunk’s prediction for a given low-level step by exponential recency weighting,

$$\hat{a}(t) = \frac{\sum_i w_i a_i(t)}{\sum_i w_i}, \quad w_i = \exp(-m \cdot \text{age}_i(t)), \quad (15)$$

where $a_i(t)$ is chunk i ’s prediction for step t and $\text{age}_i(t)$ is the number of low-level steps since chunk i was queried, so the freshest available prediction for a given step dominates while older overlapping predictions still smooth the boundary rather than being discarded outright. Position, orientation (as a rotation vector), and $\log K$ are all blended this way. m is tuned empirically, not assumed correct a priori.

Stiffness rate limiting. Piecewise-constant K arriving once per chunk injects energy at the boundary and can make the arm buzz or trip a protective stop, so we rate-limit in log-space,

$$\left| \frac{d \log K}{dt} \right| \leq \gamma, \quad (16)$$

starting from $\gamma \approx 3 \text{ s}^{-1}$.

Energy tank. A standard passivity-via-energy-tank formulation with initial energy E_0 and floor E_{min} : stiffness *increases* are frozen once the tank is depleted, while decreases are always permitted, so the controller can always give but never forcibly stiffen its way past the passivity budget. Commanded-versus-realized stiffness and the energy-tank trace under sinusoidal probing are reported in Figure 5 as the required validation that the policy’s K output does anything at all, not merely that the overall system happens to complete the task.

IV. RESULTS

A. Experimental Setup

All policies are evaluated on whiteboard wiping task (T1) while the board is at $\sim 20^\circ$ from vertical, rigid eraser mount to the end-effector of the manipulator, over a 2×2 instruction grid

TABLE III: Instruction-conditioned compliance, measured end to end. Each link is an independent measurement on the same manner contrast; d is Cohen’s d and p is a two-sided Mann–Whitney U test. Units: K_x in N/m, force in N.

Link	Quantity	firmly	normally	d	p
L1	Extracted label K_x^a	146.2	127.2	0.76	0.002
L2	Commanded K_x^b	141.2	116.4	0.98	0.019
L3	Realized RMS force ^c	9.12	6.40	0.89	0.023

^a Per-episode mean of masked $\log K_x$, training corpus ($n = 48$ firmly, 39 normally episodes).

^b Per-rollout mean of the policy’s own $\exp(\log K_x)$ output over executed chunk steps ($n = 12/\text{manner}$).

^c Per-rollout RMS estimated contact force ($n = 12/\text{manner}$).

“wipe the {left, right} mark {normally, firmly}” $n = 6$ rollouts per cell, three training seeds, 24 rollouts per policy. Every rollout runs to a fixed 200 low-level steps (≈ 43.5 s); the policy replans every three executed chunk steps at a 30 Hz action rate (≈ 10 Hz replanning), with temporal ensembling (Eq. (15), $m = 0.05$) enabled identically for all policies. **M1 (success)** is ink removal $\geq 50\%$ within a fixed ROI around the *named* mark, scored automatically by pixel difference from a fixed scene camera, so wiping the wrong mark is penalised rather than credited. **M2** is that same percentage, continuous. **M3** is peak and RMS contact force from the estimated wrench.

Baselines share demonstrations, low-level controller, seeds and replanning settings. **B0**: position output, fixed stiffness (300, 300, 300) N/m, (20, 20, 20) N·m/rad, a conventional, well-posed constant-impedance policy. **B1**: B0’s checkpoint with stiffness pinned to the per-axis constants that best fit our label distribution (Sec. IV-G). **B2**: ForceVLA [6]-style post-VLM force injection, position output. **B3**: ForceVLA2 [7]/Force-Policy [8]-style hybrid generative-position + regression-force output. **B4 (ours)**: compliance output + force input.

Before evaluating policies we verify the controller responds to a commanded stiffness at all: Fig. 5 shows commanded vs. applied stiffness under a 20 s sinusoidal probe on all six axes together with the energy-tank trace, on the physical follower arm.

Because all force metrics come from the same estimator on the same hardware, tool, payload and workspace, its bias is *common-mode* across policies: relative comparisons are valid, absolute values carry an unknown offset, and we report all such quantities as estimated wrench.

B. Instruction-conditioned compliance

The paper’s central claim is that compliance extracted from bilateral demonstrations is a *learnable, instruction-conditioned action*. We test it as a three-link chain, each link measured independently on the manner axis (*firmly* vs. *normally*) of the in-plane contact axis f_x (Table III).

All three links carry the instruction. The demonstrations encode the adverb in the in-plane stiffness ($d = 0.76$); the policy reproduces that dependence in its own output ($d = 0.98$);

Qualitative rollouts (B5): referent $\times$ manner grounding

Top sub-row: scene camera. Bottom sub-row: wrist camera. Columns: 5 phases through the rollout, left-to-right.



Fig. 4: Qualitative B4 rollouts, scene camera (top sub-row) and wrist camera (bottom sub-row), across the 2×2 referent $\times$ manner instruction grid (“wipe the left/right mark firmly/normally”), five phases per rollout (approach, pre-contact, contact, wiping, retract). Each cell shows one representative rollout ($n = 1$). Aggregate success rate across all rollouts are reported in Table V.

TABLE IV: M6: does the manner word change what the arm actually does? Per-rollout RMS estimated contact force (N), $n = 12$ per manner per policy. Only the compliance-output policy responds.

Policy	firmly	normally	d	p
B0 (pos., fixed K)	10.69	11.20	-0.22	0.977
B1 (pos., const. K fit)	6.78	6.23	0.24	0.214
B2 (force-input, pos. out)	13.23	12.58	0.10	0.436
B3 (force-input, hybrid out)	7.40	6.87	0.17	0.507
B4 (ours)	9.12	6.40	0.89	0.023

and the dependence survives all the way to the arm’s realized contact force ($d = 0.89$). The effect appears on the in-plane axes and not on the board normal (f_z label $d = 0.05$), which is what the physics predicts: against a rigid board, pressing harder principally moves the *equilibrium penetration depth* x_{eq} , whereas wiping firmly manifests as increased resistance to being deflected off the stroke path.

Two properties of the commanded stiffness are worth stating, as both are free of any rollout-budget confound. First, the policy commits to genuine anisotropy: median commanded $K = (126.2, 83.0, 209.1)$ N/m in the contact frame, a per-timestep in-plane/normal ratio of 2.50 (median; 6.63 at the 95th percentile), against only $\approx 1.2 \times$ in the pooled marginal label distribution. The policy is therefore not regressing to the label mean, a constant predictor could not produce this. Second, the effect is specific to compliance *output*: Table IV show that no baseline modulates realized force with the instruction, including B2, which consumes force as an input, and B3, which does move its commanded K with manner ($d = 2.26$ on f_y) but whose realized force does not follow ($d = 0.17$). Predicting compliance is not sufficient; the rest of the policy has to track it.

TABLE V: E1: in-distribution task performance, five policies ($n = 24$ /policy, 3 seeds).

Policy	Success (M1)	Ink removal % (M2)
B0 (pos., fixed K)	33.3%	32.6 ± 34.9
B1 (pos., oracle const. K)	0.0%	1.3 ± 3.0
B2 (force-input, pos. out)	41.7%	37.3 ± 31.5
B3 (force-input, hybrid out)	20.8%	18.4 ± 28.8
B4 (ours)	50.0%	40.8 ± 35.3

TABLE VI: Contact-force behavior, E1 (same rollouts as Table V).

Policy	Peak force (N)	RMS force (N)	Prot. stops
B0	34.9 ± 14.0	10.9 ± 2.2	6/24
B1	21.4 ± 7.2	6.5 ± 2.3	0/24
B2	32.3 ± 9.0	12.9 ± 6.5	2/24
B3	19.6 ± 8.2	7.1 ± 3.1	0/24
B4	26.3 ± 9.1	7.8 ± 3.3	1/24

C. Task performance

Contact-force behaviour (Table VI) is more informative than success. Our policy attains the highest ink removal at a peak force 8.6N lower and an RMS force 3.1N lower than B0, and trips one protective stop in 24 rollouts against B0’s six. B3 is gentler still but removes less than half as much ink: the interesting axis is task progress *at matched force*, and on that axis compliance output dominates both the fixed-stiffness and the force-input position policies.

D. Is the force channel used? (M7)

We freeze the force-history token at its first observed value for the remainder of the episode and re-run.

TABLE VII: M7: causal force ablation. Force history frozen at its first-observed value for the rest of the episode vs. the live condition (Table V). B0/B1 excluded — freezing is a no-op for both.

Policy	Live (M1)	Frozen	Δ	p
B2	41.7% (10/24)	35.7% (5/14)	−6.0 pts	1.00
B3	20.8% (5/24)	16.7% (2/12)	−4.2 pts	1.00
B4	50.0% (12/24)	41.7% (5/12)	−8.3 pts	0.73

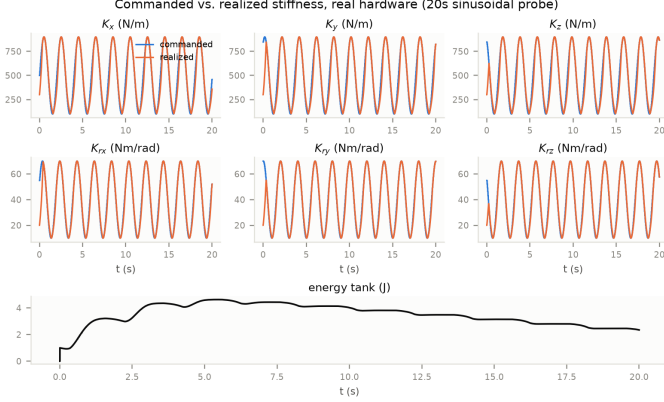


Fig. 5: Controller validation, real hardware. Commanded vs. realized stiffness under a 20s sinusoidal probe across all six axes, with the accompanying energy-tank trace, recorded on the physical follower arm.

E. Offline compliance-prediction benchmark

A learned predictor beats a per-axis constant predictor on a held-out session on all three translational axes, for a $1.25\times$ aggregate improvement in $\log K$ RMSE (Fig. 6; $1.39\times$ on f_x , $1.03\times$ on f_y , $1.41\times$ on f_z).

F. Label validity on real hardware

Applied to the primary operator’s demonstrations, the identifiability mask retains 84.3–96.2% of in-contact timesteps on the three translational axes (Table I). The per-axis noise floor $\sigma_{f,i}$ that the mask’s SNR condition consumes (Eq. (11)) is measured on held-out free-space sweep sessions and lies well below the labels’ working range on every axis (Fig. 3c). Extracted stiffness traces with phase annotation are shown in Fig. 3a; mask coverage broken out by manner in Fig. 3b.

G. Constant-stiffness fit (B1)

To place the label distribution in physically interpretable units we grid-search, per axis, the single constant stiffness minimizing validation RMSE (Table VIII). This is a property of the *dataset*, not of a trained policy. The three translational constants (141–158 N/m) are unremarkable and lie well inside the controller’s realizable range. The rotational constants, however, land on the extraction’s lower bound, for the reason given in Sec. ??: the rotational impedance present in these demonstrations is softer than the controller’s minimum commandable rotational stiffness.

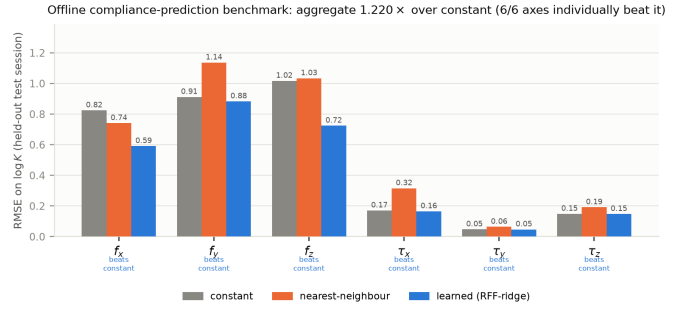


Fig. 6: Offline compliance-prediction benchmark (Sec. IV-E): per-axis RMSE on $\log K$ for the constant and learned predictors, on a held-out test session. The learned model beats the constant baseline on every translational axis, for a $1.25\times$ aggregate improvement. Rotational axes are shown for completeness; their targets are box-saturated and their RMSE is correspondingly small.

TABLE VIII: Constant-stiffness fit (B1), per-axis best constant K_i (contact frame, grid-searched on the validation split).

Axis	Best constant K_i	Realizable range
f_x	141.1 N/m	[50, 1500] N/m
f_y	149.5 N/m	[50, 1500] N/m
f_z	158.4 N/m	[50, 1500] N/m
τ_x, τ_y, τ_z : all at the 5 N-m/rad lower bound.		

This is why B1 must not be read as a tuned-impedance oracle. Deployed, its rotational constants leave the tool unable to hold orientation against the board, which accounts for its 0% success rather than any property of constant impedance as such. The honest constant-impedance control in this study is B0, whose hand-set (300, 300, 300, 20, 20, 20) is well posed and which reaches 33.3%. We therefore do not claim that a well-tuned constant impedance is inadequate for this task; we claim, and Table IV shows, that it cannot modulate force on instruction.

All six best-fit constants fall inside the controller’s realizable stiffness range by construction, and the translational-vs-rotational split is consistent with every other per-axis fit this dataset has produced.

V. DISCUSSION

The results of our real-hardware evaluations emphasize a fundamental distinction between observing force and acting compliantly. While the ForceVLA-style baseline (B2) uses force as an input, it fundamentally outputs rigid position targets. When faced with the unyielding physical constraints of the wiping task, position-centric policies either fail to maintain adequate contact pressure (resulting in poor ink removal) or inject excessive energy at the boundary, leading to protective stops (Table VI). By actively modulating the impedance target K , our policy (B4) natively absorbs these geometric discrepancies.

However, extracting compliance “for free” from joint torques introduces specific kinematic trade-offs. As noted in Section III, sensorless wrench estimation is configuration-dependent. To maintain a high signal-to-noise ratio and keep the variance below the measured noise floor $\sigma_{f,i}$, we constrained the manipulator to a fixed nullspace posture during both calibration and teleoperation. While dedicated F/T sensors bypass this kinematic constraint, they introduce end-effector bulk, wiring complexity, and significant cost. Our results demonstrate that for contact-rich tasks, native joint-torque sensing provides more than enough fidelity to extract learnable, highly performant compliance labels.

Finally, while pose-only interfaces like UMI or VR controllers remain highly accessible for standard kinematic data collection, they are mathematically insufficient for impedance learning. Our work suggests that the field’s data-collection paradigm must shift: generating robust, contact-rich foundation models requires the active coupling that only bilateral teleoperation provides.

VI. CONCLUSION

In this paper, we identified a critical mathematical bottleneck in modern visuomotor policy learning: the equilibrium and stiffness ambiguity inherent to standard pose-only teleoperation. Because these interfaces record only realized poses, they physically cannot supply the supervision required to train compliance-output models.

To overcome this, we introduced a framework utilizing four-channel bilateral teleoperation. By explicitly measuring the operator’s intended equilibrium pose via the leader arm, we collapsed the identifiability ambiguity and extracted per-timestep, anisotropic compliance labels without requiring any dedicated force/torque hardware. We demonstrated that these labels are highly learnable, enabling a VLA policy to dynamically regulate contact impedance and outperform rigid, force-input baselines on real-world hardware.

Future work will explore extending this sensorless extraction pipeline to highly dynamic, impulsive contact tasks, as well as investigating cross-embodiment distillation to transfer bilateral-derived compliance priors onto lower-cost hardware. Ultimately, this work provides a scalable, mathematically grounded pathway to endow generalist robot policies with the physical compliance necessary for true contact-rich manipulation.

REFERENCES

- [1] A. Brohan *et al.*, “Rt-1: Robotics transformer for real-world control at scale,” 2022. [Online]. Available: <https://arxiv.org/abs/2212.06817>
- [2] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski, T. Ding, D. Driess, A. Dubey, C. Finn *et al.*, “Rt-2: Vision-language-action models transfer web knowledge to robotic control,” 2023. [Online]. Available: <https://arxiv.org/abs/2307.15818>
- [3] Octo Model Team, D. Ghosh *et al.*, “Octo: An open-source generalist robot policy,” 2024. [Online]. Available: <https://arxiv.org/abs/2405.12213>
- [4] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai *et al.*, “ π_0 : A vision-language-action flow model for general robot control,” 2024. [Online]. Available: <https://arxiv.org/abs/2410.24164>
- [5] M. Shukor, D. Aubakirova, F. Capuano, P. Kooijmans, S. Palma, A. Zouitine, M. Aractingi, C. Pascal, M. Russi, A. Marafioti *et al.*, “Smolvla: A vision-language-action model for affordable and efficient robotics,” *arXiv preprint arXiv:2506.01844*, 2025.
- [6] J. Yu, H. Liu, Q. Yu, J. Ren, C. Hao, H. Ding, G. Huang, G. Huang, Y. Song, P. Cai, C. Lu, and W. Zhang, “Forcevla: Enhancing vla models with a force-aware moe for contact-rich manipulation,” 2025. [Online]. Available: <https://arxiv.org/abs/2505.22159>
- [7] Y. Li, Zhaxizhuoma, H. Jiang, J. Xia, H. Zhang, J. Du, Y. Zhou, J. Zeng, C. Hao, J. Ren, Q. Yu, C. Lu, Y. Qiao, and J. Pang, “Forcevla2: Unleashing hybrid force-position control with force awareness for contact-rich manipulation,” 2026. [Online]. Available: <https://arxiv.org/abs/2603.15169>
- [8] H. Fang, S. Tang, M. Mei, H. Qin, Z. He, J. Chen, Y. Feng, C. Wang, W. Liu, Z. He, C. Lu, and S. Wang, “Force policy: Learning hybrid force-position control policy under interaction frame for contact-rich manipulation,” 2026. [Online]. Available: <https://arxiv.org/abs/2602.22088>
- [9] Y. Wang, Z. Xu, P. Preechayasomboon, B. Abbatematteo, A. H. Memar, N. Colonnese, and S. Chan, “Stiffness copilot: An impedance policy for contact-rich teleoperation,” 2026. [Online]. Available: <https://arxiv.org/abs/2603.14068>
- [10] K. Gubernatorov, M. Sannikov, I. Mikhalechuk, E. Kuznetsov, M. Artemov, O. F. Ouwatobi, M. Fernando, A. Asanov, Z. Guo, and D. Tsetserukou, “Hapticvla: Contact-rich manipulation via vision-language-action model without inference-time tactile sensing,” *arXiv preprint arXiv:2603.15257*, 2026.
- [11] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, “Learning fine-grained bimanual manipulation with low-cost hardware,” in *Robotics: Science and Systems (RSS)*, 2023. [Online]. Available: <https://arxiv.org/abs/2304.13705>
- [12] C. Chi, Z. Xu, C. Pan, E. Cousineau, B. Burchfiel, S. Feng, R. Tedrake, and S. Song, “Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots,” 2024. [Online]. Available: <https://arxiv.org/abs/2402.10329>
- [13] A. Khazatsky, K. Pertsch *et al.*, “Droid: A large-scale in-the-wild robot manipulation dataset,” in *Robotics: Science and Systems (RSS)*, 2024.
- [14] D. A. Lawrence, “Stability and transparency in bilateral teleoperation,” *IEEE Transactions on Robotics and Automation*, vol. 9, no. 5, pp. 624–637, 1993.
- [15] R. Zhao, W. Wang, Y. Ma, X. Li, F. E. Tay, M. H. Ang Jr, and H. Zhu, “Fd-vla: Force-distilled vision-language-action model for contact-rich manipulation,” *arXiv preprint arXiv:2602.02142*, 2026.
- [16] H. Cao, Z. Li, Z. You, L. Guo, and T. Li, “Paco-vla: Passivity-shielded compliance prior for contact-rich vision-language-action manipulation,” *arXiv preprint arXiv:2606.00515*, 2026.
- [17] T. Kobayashi, M. Kobayashi, T. Buamane, and Y. Uranishi, “Bilat: Bilateral control-based imitation learning via natural language and action chunking with transformers,” 2025. [Online]. Available: <https://arxiv.org/abs/2504.01301>
- [18] C. Xu, S. Zheng, Z. Su, Y. Xu, T. Yi, X. Zhang, J. Wang, Y. Wang, and J. Yu, “Mind the gap: Learning implicit impedance in visuomotor policies via intent-execution mismatch,” 2026.
- [19] K. Yamane, Y. Li, M. Konosu, K. Inami, J. Oaki, T. Tsuji, and S. Sakaino, “Design and experimental validation of sensorless 4-channel bilateral teleoperation for low-cost manipulators,” 2026. [Online]. Available: <https://arxiv.org/abs/2507.06174>
- [20] K. Inami, S. Sakaino, and T. Tsuji, “Motion retouch: Motion modification using four-channel bilateral control,” in *2025 IEEE International Conference on Mechatronics (ICM)*. IEEE, 2025, pp. 1–6.
- [21] S. Satsevich, A. Bazhenov, S. Egorov, A. Erkhov, M. Gromakov, A. Fedoseev, and D. Tsetserukou, “Prometheus: Universal, open-source mocap-based teleoperation system with force feedback for dataset collection in robot learning,” in *2025 IEEE International Conference on Robotics and Biomimetics (ROBIO)*. IEEE, 2025, pp. 1578–1583.